AI Use Declaration Form

Course: Introduction to Artificial Intelligence
Lab Title: Lab 2 — Predictive Analytics with Machine Learning
Student Name: __James Emmanuel Yeboah-Asiamah____
Student ID: ____14022028______
GitHub Repository Link: ______https://github.com/JamesEmmanuel20/lab-2-predictive-analytics.git_____
Date Submitted: _________________3rd June 2026____

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	**Yes**
Estimated percentage of the work influenced by AI	12%
Did you attach evidence of AI use where applicable?	**Yes**

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Gemini	Explanation of why I obtained an error for	ValueError Traceback (most recent call last)Cell In[44], line 14 10 caec_calc_order = ['no', 'Sometimes', 'Frequently', 'Always'] 11 # TODO (optional but encouraged): engineer a domain feature, e.g. BMI = Weight / Height**2. 12 # Discuss whether including BMI makes the task "too easy" / leaks the target. 13 ordinal_encoder = OrdinalEncoder(categories=[caec_calc_order, caec_calc_order], handle_unknown='use_encoded_value', unknown_value= -1)---> 14 obesity[['CAEC', 'CALC']] = ordinal_encoder.fit_transform(obesity[['CAEC', 'CALC']]) 15 16 obesity = pd.get_dummies(obesity, columns = ['Gender', 'MTRANS'], drop_first = True) 17 File	Part 2.2, Part 2.4, Part	I verified it by implementing the changes then running the cell

		c:\Users\james\OneDrive\Desktop\JAMES_SCHOOL\25-26 ACADEMIC YEAR\SEMESTER 2\Introduction to AI\ai_labs\second_ai_lab\.venv\Lib\site-packages\sklearn\utils_set_output.py:319, in _wrap_method_output.<locals>.wrapped(self, X, *args, **kwargs) 317 @wraps(f) 318 def wrapped(self, X, *args, **kwargs):--> 319 data_to_wrap = f(self, X, *args, **kwargs) 320 if isinstance(data_to_wrap, tuple): 321 # only wrap the first output for cross decomposition 322 return_tuple = (323 _wrap_data_with_container(method, data_to_wrap[0], X, self), 324 *data_to_wrap[1:], 325)File c:\Users\james\OneDrive\Desktop\JAMES_SCHOOL\25-26 ACADEMIC YEAR\SEMESTER 2\Introduction to AI\ai_labs\second_ai_lab\.venv\Lib\site-packages\sklearn\base.py:969, in TransformerMixin.fit_transform(self, X, y, **fit_params) 954 warnings.warn(... 130 " which is incompatible with values of type" 131 f" '{type(Xi[0]).__name__}'." 132)ValueError: could not convert string to float: 'no' why am I getting this error and how do I resolve it,		

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	N/A	
Prompt history	N/A	

<table>
<tr>
<td>Screenshot of AI interaction</td>
<td>Yes</td>
<td>

'Always'] 11 # TODO (optional but encourage
domain feature, e.g. BMI = Weight / Height**2

Ah, the `ValueError: could not convert string to float: 'no'`. Thi
learn hiccup when dealing with categorical encodings.

Why you are getting this error

By default, scikit-learn's `OrdinalEncoder` expects the categories you pass
`categories` argument to match the **exact data type** found in your DataFr

Looking closely at your dataset (this looks like the UCI Obesity dataset), the
column likely contains a mix of actual string values (like `'Sometimes'`) and
values that Python interpreted as actual booleans (`False` / `True`) or float
loaded.

When pandas reads a CSV, it often automatically converts the string `'no'` i
`False`. So, while your list says `['no', 'Sometimes'...]`, scikit-learn loo
DataFrame, sees a boolean `False`, and when it tries to map it to your list o

How to resolve it

You can fix this easily by forcing the `CAEC` and `CALC` columns to be explicitly
(`str`) right before you pass them into the encoder.

Update your code to look like this:

Python

<pre>
caec_calc_order = ['no', 'Sometimes', 'Frequently', 'Always']

1. Force both columns to be string type to prevent pandas boolean
obesity['CAEC'] = obesity['CAEC'].astype(str)
obesity['CALC'] = obesity['CALC'].astype(str)
</pre>

</td>
</tr>
</table>

Why this happens

This error usually triggers because of one of two reasons:

1. **You already ran this cell once:** When you run `pd.get_dummies(..` `drop_first=True)`, pandas deletes the original columns (`Gender` replaces them with the new dummy columns (like `Gender_Male`, `MTRANS_Public_Transportation`). If you execute the same Jupy **second time** without reloading the data, the original columns are g `KeyError`.
2. **They were dropped or renamed earlier:** The columns might have b named differently (e.g., lowercased as `'gender'` or `'mtrans'`) ir

How to resolve it

Step 1: Reload your dataset

If you just accidentally re-ran the cell, go back up to the top of your noteb read the CSV (e.g., `obesity = pd.read_csv(...)`) and run the cells se

Here is a breakdown of the most important parameters, categorized by wl

1. Core Ensemble Parameters (Forest Structure)

These parameters control the size of the forest and how data is sampled.

- `n_estimators` (default= `100`)
  - **What it represents:** The number of decision trees in the forest
  - **Impact:** Generally, more trees increase model performance an they smooth out predictions, but after a certain point, you hit c and it will slow down training time.

		## 2. Tree Splitting Parameters (Feature Control) These parameters control how each individual tree selects features to split on. ○ `max_features` (default= `"sqrt"`) ○ **What it represents:** The number of features to consider when looking for the split at each node. ○ **Options:** Can be an integer, a float (percentage of features), `"sqrt"` (square of total features), or `"log2"` . ○ **Impact:** Reducing this number forces trees to be more diverse (less correlatec each other), which reduces overfitting but can increase bias if it's set too low. ○ `criterion` (default= `"gini"`) ○ **What it represents:** The function used to measure the quality of a split. ○ **Options:** `"gini"` (Gini impurity) or `"entropy"` / `"log_loss"` (Information ○ **Impact:** Gini is slightly faster to compute, while entropy sometimes results in s how do you create a confusion matrix??? Brie A **confusion matrix** is a simple grid used to evaluate how well a machine learning classific model is performing. It compares the model's predicted labels against the actual true labe Here is how you create one in 3 brief steps: **1. Set Up the Grid** For a binary classification (e.g., Yes/No, Cat/Dog), draw a 2×2 grid. ○ Label the rows as **Actual** values. ○ Label the columns as **Predicted** values. 2. Fill in the Four Quadrants
Exported AI conversation	N/A	
AI-generated code snippet	N/A	
Revised version showing	N/A	

student input		

4. Student Declaration

I declare that:

Declaration Statement	**Response (Yes/No)**
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature: ___________JEmmanuel_______________________________

Date: ___________3rd June 2026_______________________________